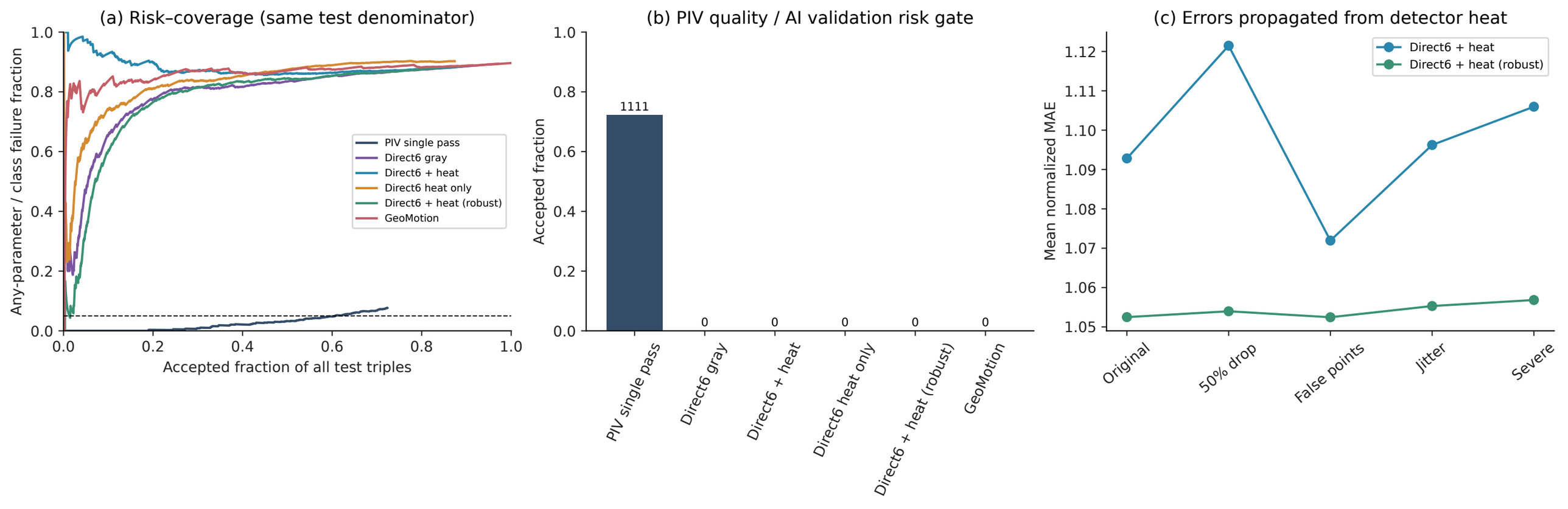
(a) Risk–coverage (same test denominator)
Any-parameter / class failure fraction
Accepted fraction of all test triples
PIV single pass
Direct6 gray
Direct6 + heat
Direct6 heat only
Direct6 + heat (robust)
GeoMotion
(b) PIV quality / AI validation risk gate
Accepted fraction
1111
0
PIV single pass
Direct6 gray
Direct6 + heat
Direct6 heat only
Direct6 + heat (robust)
GeoMotion
(c) Errors propagated from detector heat
Mean normalized MAE
Direct6 + heat
Direct6 + heat (robust)
Original
50% drop
False points
Jitter
Severe